

Multiple Choice (10 marks)

1. Statement 1| RoBERTa pretrains on a corpus that is approximate 10 x larger than the corpus BERT pretrained on. Statement 2| ResNeXts in 2018 usually used tanh activation functions.
 - (a) True, True
 - (b) False, False
 - (c) True, False
 - (d) False, True
2. Which of the following tasks can be best solved using Clustering.
 - (a) Predicting the amount of rainfall based on various cues
 - (b) Detecting fraudulent credit card transactions
 - (c) Training a robot to solve a maze
 - (d) All of the above
3. Statement 1| Word 2 Vec parameters were not initialized using a Restricted Boltzman Machine. Statement 2| The tanh function is a nonlinear activation function.
 - (a) True, True
 - (b) False, False
 - (c) True, False
 - (d) False, True
4. Statement 1| The BLEU metric uses precision, while the ROGUE metric uses recall. Statement 2| Hidden markov models were frequently used to model English sentences.
 - (a) True, True
 - (b) False, False
 - (c) True, False
 - (d) False, True
5. The disadvantage of Grid search is
 - (a) It can not be applied to non-differentiable functions.
 - (b) It can not be applied to non-continuous functions.
 - (c) It is hard to implement.
 - (d) It runs reasonably slow for multiple linear regression.

True / False (10 marks)

Answer True or False

6. True or False: The computational complexity of gradient descent is linear in the number of data points, N .
7. True or False: Using an L_2 norm regularization penalty will result in some coefficients of w being exactly zeroed out.
8. True or False: Residual connections are utilized in both ResNets and Transformers to improve training and performance of deep networks.
9. True or False: The models obtained from all selection methods, including best-subset and forward stepwise, will always be identical.
10. True or False: To achieve the same projection as SVD in PCA, the data must be transformed to have a zero mean.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Write a function to find the area of a rombus.
12. Write a function to convert all possible convertible elements in the list to float.

End of Paper